

1. (a)  $(1.5 \times 1000 \text{ kg}) \times 4200 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1} \times (80 - 60) \text{ }^{\circ}\text{C} \times (1 - 15\%)$   
 $= 1.07 \times 10^8 \text{ J}$

1M+1M  
1A 3

(b)  $1.07 \times 10^8 \text{ J} \div (4.5 \text{ kW}) \div 3600 \text{ s}$   
 $= 6.61 \text{ (hours)}$

1M  
1A 2

(c) Rate of heat transfer drops as the water temperature drops / the room temperature increases / temperature difference drops.

1A 1

1. (a) 5 min. (or 300 s)

1A

1

(b) When the heater is switched off, its temperature is higher than the metal.

1A

Or Heat continues to conduct to the metal until the same temperature is attained.

1A

Or It takes time for them to attain the same temperature.

1A

1

(c) (i)  $mc\Delta T = IVt$   
 $(0.80) c (45 - 20) = (4.0)(12)(5 \times 60)$   
 $c = 720 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$

1M

1A

2

(ii) Experimental result is higher than the actual value.  
 Not all the energy supplied by the heater goes to the metal.

1A

1A

Or Some energy is absorbed by the heater itself and / or by the thermometer.

1A

Or Heat loss to the surroundings.

1A

2

(d) Glass is not a good conductor of heat.

1A

Or Prolonged time is required for the whole glass block to attain uniform temperature.

1A

Or No insulating material is perfect, so heat loss is appreciable.

1A

1

|    |     |                                                                                                                                     |    |  |
|----|-----|-------------------------------------------------------------------------------------------------------------------------------------|----|--|
| 1. | (a) | - Put the sphere into the water bath for a few minutes                                                                              | 1A |  |
|    |     | - Put / transfer the sphere into the polystyrene cup (of water)                                                                     | 1A |  |
|    |     | - Measure the final / maximum temperature $T_f$ of the water with a thermometer                                                     | 1A |  |
|    |     |                                                                                                                                     |    |  |
|    |     | $0.80 \times c_b \times (80 - T_f) = 0.50 \times 4200 \times (T_f - T_0)$                                                           | 1A |  |
|    |     | $c_b = 2625 \times \frac{T_f - T_0}{80 - T_f} \text{ (J kg}^{-1} \text{ } ^\circ\text{C}^{-1}\text{)}$                              |    |  |
|    |     | Precautions:                                                                                                                        |    |  |
|    |     | - Dry the sphere with the towel quickly before putting it into the cup                                                              | 1A |  |
|    |     | - Make sure the sphere is fully immersed in water                                                                                   | 1A |  |
|    |     | - Stir the water thoroughly                                                                                                         | 1A |  |
|    |     |                                                                                                                                     | 6  |  |
|    | (b) | Thermal energy / heat is lost during the transfer / drying of the sphere                                                            | 1A |  |
|    |     | Or Thermal energy / heat absorbed by thermometer, stirrer or cup                                                                    | 1A |  |
|    |     | Or The temperature of the sphere is higher than $T_f$ when this final temperature is measured (i.e. $T_f$ not reaching its maximum) | 1A |  |
|    |     | Thus temperature rise of water in the cup is lower than it should be.                                                               | 1A |  |
|    |     |                                                                                                                                     | 2  |  |

| Solution |     | Marks                                                           | Remarks |
|----------|-----|-----------------------------------------------------------------|---------|
| 1.       | (a) | Amount of energy required $E = mc\Delta T$                      |         |
|          |     | $= 6 \times 4200 \times (50 - 15)$                              |         |
|          |     | $= 882000 \text{ J (or 882 kJ)}$                                |         |
|          |     |                                                                 |         |
|          |     | Power $P = \frac{E}{t} = \frac{882000}{60}$                     |         |
|          |     | $= 14700 \text{ W (or 14.7 kW)}$                                |         |
|          |     |                                                                 |         |
|          |     |                                                                 |         |
|          |     |                                                                 |         |
|          |     | 3                                                               |         |
|          | (b) | Let $m$ kg per minute be the water flow rate                    |         |
|          |     | $mc\Delta T = Pt$                                               |         |
|          |     | $m(4200)(40 - 15) = 14700 \times 60$                            |         |
|          |     | $m = 8.4 \text{ (kg min}^{-1} \text{ or kg)}$                   |         |
|          |     |                                                                 |         |
|          |     |                                                                 |         |
|          |     | 2                                                               |         |
|          | (a) | $n = \frac{pV}{RT} \propto pV \text{ (for constant } T\text{)}$ |         |
|          |     | $\frac{n_A}{n_B} = \frac{(p)(3V)}{(2p)(2V)}$                    |         |
|          |     | $n_A = 0.75 \times 0.8 \text{ mol}$                             |         |
|          |     | $= 0.6 \text{ (mol)}$                                           |         |
|          |     | 1A                                                              |         |

|    |       |                                                                                                                                                                                                         |       |                                                                                                                       |
|----|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------|
| 1. | (a)   | $5 \times 0.02 \times 3300 \times (T - 4) = 0.60 \times 4200 \times (96 - T)$<br>$T = 85.347368 \text{ }^{\circ}\text{C} \approx 85.3 \text{ }^{\circ}\text{C}$                                         | 1M    | Accept 85.0 °C to 85.4 °C<br>[Marking Remark (MR): temperature difference must be correct & accept missing '5' - 1 M] |
|    |       |                                                                                                                                                                                                         | 1A    |                                                                                                                       |
|    | (b)   | (i)                                                                                                                                                                                                     |       | 2                                                                                                                     |
|    |       |                                                                                                                                                                                                         | 1A    | [MR: Key words - heat loss to the surroundings (from container or through evaporation)]                               |
|    |       | (ii)                                                                                                                                                                                                    |       | 1                                                                                                                     |
|    |       |                                                                                                                                                                                                         | 1M+1M | Assume :                                                                                                              |
|    | (iii) | Smaller than 9 °C.<br>As the temperature of (the container with) soup drops, the rate of heat loss to the surroundings becomes lower due to a smaller temperature difference (w.r.t. the surroundings). | 1A    | Power of the heater = Rate of heat loss (of the container with soup) to the surroundings                              |
|    |       |                                                                                                                                                                                                         |       | 3 [MR: $P \times t$ ✓ + 1 term ✓ - 1M                                                                                 |
|    |       |                                                                                                                                                                                                         |       | $P \times t$ ✓ + 2 term ✓ - 2M                                                                                        |
|    |       |                                                                                                                                                                                                         |       | $P \times t$ : accept $P \times 10$                                                                                   |
|    |       |                                                                                                                                                                                                         |       | $\Delta T$ : accept (96 - 9)]                                                                                         |
|    |       |                                                                                                                                                                                                         | 1A    | [2 A marks are independent                                                                                            |
|    |       |                                                                                                                                                                                                         | 1A    | Key <b>concept</b> : smaller temperature difference]                                                                  |
|    |       |                                                                                                                                                                                                         |       | 2                                                                                                                     |

|         |                                                                                                                                                                                                                                                                                                                                                    |    |  |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|--|
| 1. (a)  | Heat given out by iron cube = Heat gained by water<br>$0.2 \times 450 \times (T - 33) = 0.6 \times 4200 \times (33 - 25)$<br>$T = 257 (^{\circ}\text{C})$                                                                                                                                                                                          | 2M |  |
|         |                                                                                                                                                                                                                                                                                                                                                    | 1A |  |
|         |                                                                                                                                                                                                                                                                                                                                                    | 3  |  |
|         |                                                                                                                                                                                                                                                                                                                                                    |    |  |
|         |                                                                                                                                                                                                                                                                                                                                                    |    |  |
| (b)     | Some water might become steam (immediately).<br><u>Or</u><br>Some water in the cup might splash.                                                                                                                                                                                                                                                   | 1A |  |
|         |                                                                                                                                                                                                                                                                                                                                                    | 1A |  |
|         |                                                                                                                                                                                                                                                                                                                                                    | 1  |  |
| (c)     | It is because some heat is lost to the surroundings through vaporization of water / as latent heat.<br><u>Or</u><br>It is because some heat is lost to the surroundings (or polystyrene cup) / The water might not be well stirred, otherwise, the water should have reached a temperature higher than $33^{\circ}\text{C}$ (i.e. underestimated). | 1A |  |
|         |                                                                                                                                                                                                                                                                                                                                                    | 1A |  |
|         |                                                                                                                                                                                                                                                                                                                                                    | 1A |  |
|         |                                                                                                                                                                                                                                                                                                                                                    | 1A |  |
|         |                                                                                                                                                                                                                                                                                                                                                    | 2  |  |
| (d) (i) | Poor thermal contact between the glass bob of the thermometer and the iron cube.                                                                                                                                                                                                                                                                   | 1A |  |
|         |                                                                                                                                                                                                                                                                                                                                                    | 1  |  |
|         |                                                                                                                                                                                                                                                                                                                                                    |    |  |
|         |                                                                                                                                                                                                                                                                                                                                                    |    |  |
| (ii)    | Infra-red (IR) thermometer.                                                                                                                                                                                                                                                                                                                        | 1A |  |
|         |                                                                                                                                                                                                                                                                                                                                                    | 1  |  |

|    |                                                                                                                                                         |    |  |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------|----|--|
| 1. | (a) The glass cup absorbs thermal energy (from the hot water) /<br>Heat is lost to the glass cup.                                                       | 1A |  |
|    |                                                                                                                                                         | 1  |  |
|    | (b) $4200 \times 0.4 \times (100 - 93) = H \times (93 - 23)$<br>$H = 168 \text{ (J } ^\circ\text{C}^{-1}) \approx 170 \text{ (J } ^\circ\text{C}^{-1})$ | 1M |  |
|    |                                                                                                                                                         | 1A |  |
|    | (c) (i) Infra-red (radiation) / thermal radiation                                                                                                       | 2  |  |
|    |                                                                                                                                                         | 1A |  |
|    |                                                                                                                                                         | 1  |  |
|    |                                                                                                                                                         | 1A |  |
|    |                                                                                                                                                         | 1  |  |
|    | (ii) Remote control / infra-red (IR) thermometer /<br>night vision camera                                                                               | 1A |  |
|    |                                                                                                                                                         | 1  |  |